Ain Shams University

Faculty of Engineering

CHEP – Communications Systems Engineering Program

Spring 2026 | ECE359: Digital Image Processing & Multimedia

Sheet 3

Intensity Transformations, Histograms & Spatial Filtering

Part A Textbook Problems

1. [1] (Constructing Exponential Transformations)

Exponential functions of the form $e^{-\alpha r^2}$ (with $\alpha > 0$) can be used to construct smooth intensity transformation functions $s = T(r)$ defined on $[0,\ L-1]$. Construct a transformation function matching each of the three curve shapes below. The labelled constants are *input* parameters and must appear explicitly in your formula. (L_0 is not required in curve (c).)

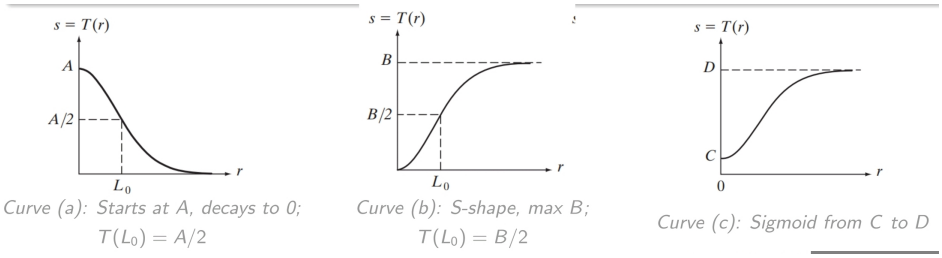

Figure 1: Curve shapes (a), (b), and (c).

(a) A curve starting at $s = A$ when $r = 0$ that decays monotonically to zero, with $T(L_0) = A/2$.
(b) An S-shaped curve with maximum output B, satisfying $T(L_0) = B/2$, constructed by reversing and vertically shifting curve (a).
(c) A curve that rises smoothly from C to D across the full input range $[0,\ L-1]$.

For each case, express any free constants in terms of A, B, C, D, L_0, and L.

[1] R.C. Gonzalez & R.E. Woods, *Digital Image Processing*, 4th ed. (Pearson, 2018), Problem 3.2.

2. [2] **(Bit-Plane Slicing via Intensity Transformations)**

Propose a complete set of intensity-slicing transformations $T_k(r)$ capable of extracting every individual bit plane of an 8-bit monochrome image.

Recall that each pixel value $r \in \{0, 1, \ldots, 255\}$ has 8 binary bits $b_7 b_6 \cdots b_1 b_0$ where b_7 is the MSB, and bit-plane k isolates bit b_k of every pixel.

As a reference, the transformation that extracts the MSB (plane 7) is:

$$T_7(r) = \begin{cases} 0 & 0 \leq r \leq 127 \\ 255 & 128 \leq r \leq 255 \end{cases}$$

(a) Following the example above, derive the general transformation $T_k(r)$ for bit plane k ($k = 0, 1, \ldots, 7$). Express all intensity interval boundaries in terms of k.
(b) How many distinct intervals does $T_k(r)$ require for bit plane k? Justify your answer.
(c) Briefly explain the practical use of bit-plane slicing in image compression and steganography.

3. [3] **(Bit Planes & Histograms)**

(a) What effect does **setting the lower-order bit planes to zero** have on the histogram of an image in general? Describe the change in the set of possible pixel values and explain how it manifests as spikes or spread in the histogram.
(b) What would be the effect on the histogram if we instead **set the higher-order bit planes to zero**?

4. [4] **(Why Discrete HE Does Not Yield a Flat Histogram)**

In the continuous case, the transformation $s = T(r) = (L-1) \int_0^r p_r(w)\, dw$ guarantees a uniform PDF for s.

(a) Explain why discrete histogram equalisation does **not**, in general, yield a perfectly flat histogram. Identify the specific step that breaks the uniformity guarantee.
(b) What property of the rounding step causes multiple input levels to collapse onto a single output level?
(c) Under what special condition on the original histogram would discrete HE produce an exactly flat result?

[2] Gonzalez & Woods, 4th ed., Problem 3.4.
[3] Gonzalez & Woods, 4th ed., Problem 3.5.
[4] Gonzalez & Woods, 4th ed., Problem 3.6.

5. [5] **(Histograms of Combined Images)**

Two images $f(x, y)$ and $g(x, y)$ have histograms h_f and h_g. Give the conditions under which the histogram of each result below can be predicted in terms of h_f and h_g, and explain how to obtain it in each case.

(a) $f(x, y) + g(x, y)$
(b) $f(x, y) - g(x, y)$
(c) $f(x, y) \times g(x, y)$
(d) $f(x, y) \div g(x, y)$

(*Note: Histograms carry no spatial information. Consider the special case where g is a constant image, i.e. every pixel equals a constant c, and describe the effect on the histogram in each case.*)

Part B Section Problems

6. (Intensity Transformation Matching)

Four distinct pointwise intensity transformations are applied independently to the same original grayscale image (shown below), producing four output images labelled A, B, C, and D. Four transformation curves (1)–(4) are also shown.

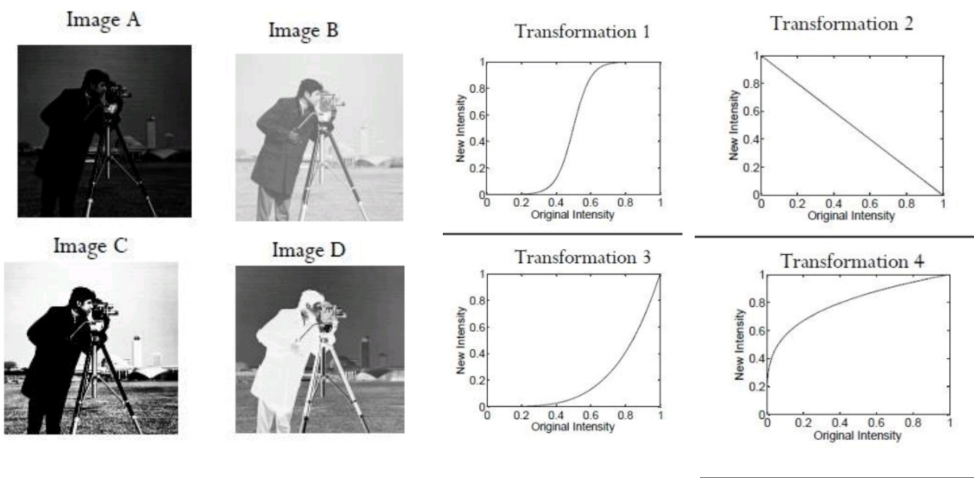

Figure 2: Transformed images A–D and transformation curves 1–4.

Match each transformed image (A, B, C, D) to the correct transformation curve (1, 2, 3, 4) and state your reasoning for each match.

[5] Gonzalez & Woods, 4th ed., Problem 3.13.

7. (Histogram Matching — Identify the Operation)

The coin image and its histogram are shown below.

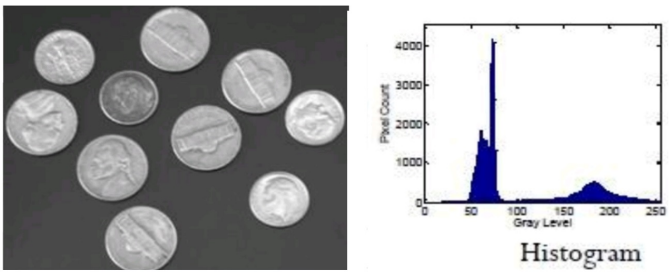

Figure 3: Coin image and its original histogram.

Four histogram-modifying operations are listed below. Match each operation to the correct modified histogram (a)–(d) shown, and state your reasoning.

(1) Increase brightness (state the approximate increase in intensity level)
(2) Decrease brightness (state the approximate decrease)
(3) Increase contrast
(4) Decrease contrast

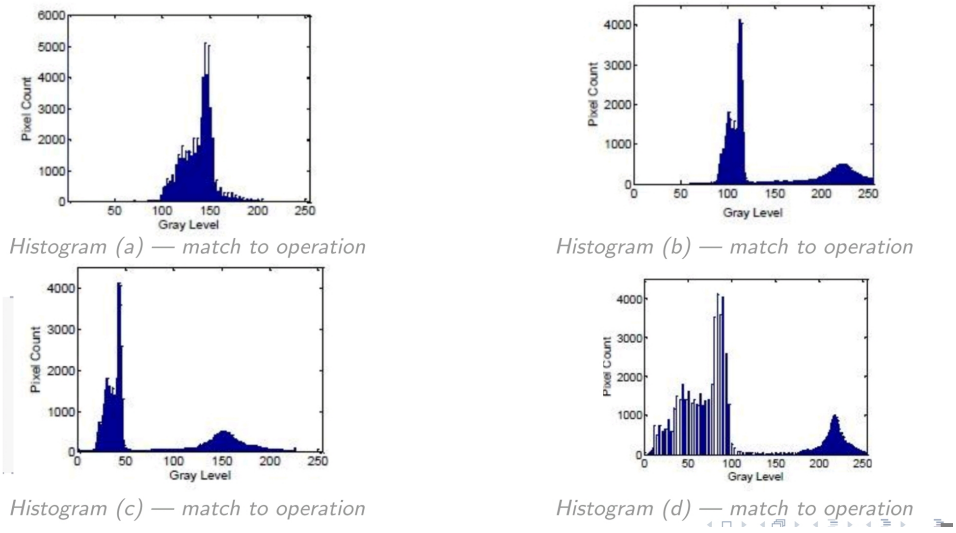

Figure 4: Modified histograms (a) through (d).

8. [6] **(Continuous Histogram Equalisation — Illustration)**

Suppose that the (continuous) intensity values in an image have the following probability density function (PDF):

$$p_r(r) = \begin{cases} \dfrac{2r}{(L-1)^2} & \text{for } 0 \le r \le L-1 \\ 0 & \text{otherwise} \end{cases}$$

(a) Derive the histogram equalisation transformation function

$$s = T(r) = (L-1)\int_0^r p_r(w)\,dw$$

and simplify the result.

(b) Using the change-of-variables theorem $p_s(s) = p_r(r)\left|\dfrac{dr}{ds}\right|$, compute the PDF of the output s and verify that it is uniform over $[0,\, L-1]$.

(c) State the general principle illustrated by this example: why does the continuous HE transformation guarantee a uniform output PDF?

9. [7] **(Discrete Histogram Equalisation — Step-by-Step)**

A 3-bit image ($L = 8$) of size 64×64 pixels ($MN = 4096$) has the intensity distribution given in Table 3.1 below, where intensity levels are integers in $[0,\, L-1] = [0,\, 7]$.

Intensity Level r_k	**Pixel Count** n_k	$p_r(r_k) = n_k/MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

(a) Compute the cumulative distribution function (CDF) at each intensity level.

(b) Apply the discrete HE transformation

$$s_k = (L-1)\sum_{j=0}^{k} p_r(r_j)$$

to obtain the mapped output levels. Round each s_k to the nearest integer in $[0,\, 7]$.

(c) Construct the equalised histogram by summing the pixel counts of all input levels that map to the same output level.

(d) Comment on why the resulting histogram is approximately — but not perfectly — uniform.

[6] R.C. Gonzalez & R.E. Woods, *Digital Image Processing*, 4th ed. (Pearson, 2018), Example 3.4.
[7] Gonzalez & Woods, 4th ed., Example 3.5.

10. (Continuous Histogram Specification)

An image has the input intensity PDF:

$$p_r(r) = \begin{cases} \dfrac{3r^2}{(L-1)^3} & 0 < r < L-1 \\ 0 & \text{otherwise} \end{cases}$$

It is desired to transform the image so that its output intensities follow the target PDF:

$$p_z(z) = \begin{cases} \dfrac{2}{L-1} - \dfrac{2z}{(L-1)^2} & 0 \le z \le L-1 \\ 0 & \text{otherwise} \end{cases}$$

(a) Compute the input equalisation transform $S = T(r) = (L-1)\displaystyle\int_0^r p_r(w)\,dw$ and simplify.

(b) Compute the target cumulative distribution $G(z) = (L-1)\displaystyle\int_0^z p_z(t)\,dt$ and simplify.

(c) Set $G(z) = S$ and solve for z as a function of r. Apply the quadratic formula and justify which root is physically valid (i.e. lies within $[0,\ L-1]$).

(d) Write the final closed-form mapping $z = f(r)$ explicitly.

11. (Discrete Histogram Specification) *[Student Task]*

Access interactive tool via the clickable link: **W3 — Histogram Processing Task**

Choose any values for the pixel counts n_k of a 3-bit image ($L = 8$, so $k = 0, 1, \ldots, 7$), ensuring the total $MN = \sum n_k$ is a positive integer. Then perform the following steps **manually**, and verify your result using the tool.

Step 1. Construct the input histogram. List your chosen n_k values and compute $p_r(r_k) = n_k/MN$ for each level.

Step 2. Compute the CDF. $\text{CDF}(k) = \displaystyle\sum_{j=0}^{k} p_r(r_j)$ for $k = 0, \ldots, 7$.

Step 3. Define the target histogram. Choose a desired output distribution $p_z(z_k)$ (e.g. uniform, or any shape of your choice) and compute its CDF: $G(z_k) = \displaystyle\sum_{j=0}^{k} p_z(z_j)$.

Step 4. Apply the specification mapping. For each input level r_k, find the output level z_m such that $G(z_m)$ is closest to $\text{CDF}(k)$. This gives the mapping $r_k \mapsto z_m$.

Step 5. Construct the output histogram. Assign pixel counts n_k to the mapped output levels and plot the result.

Compare your manual result with the tool output and comment on any discrepancies.